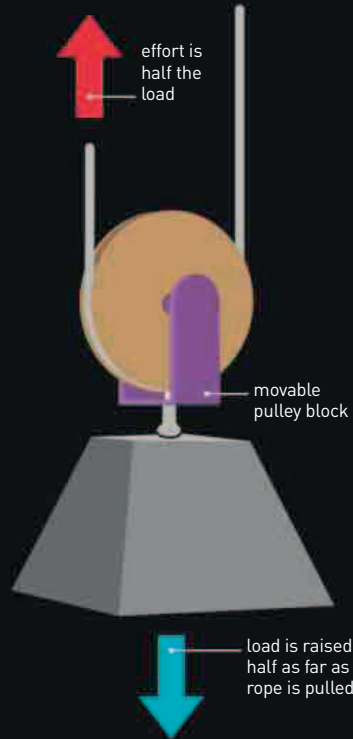
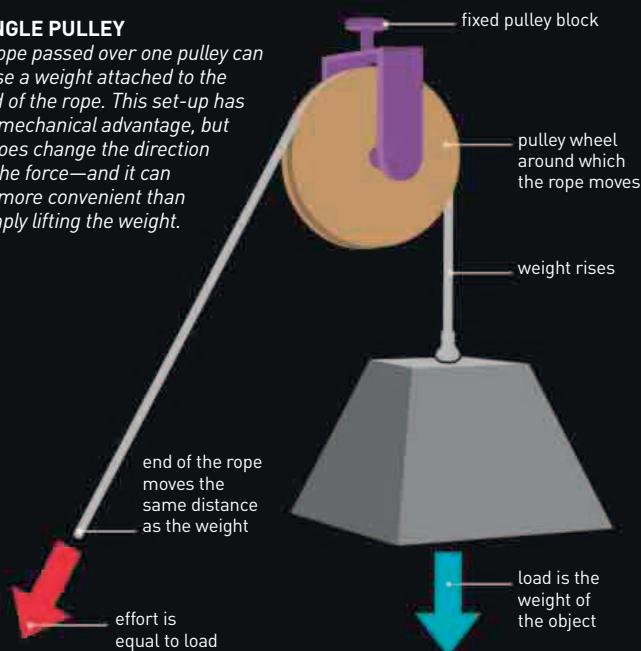


PULLEYS

A simple pulley—a rope passed over a free-moving wheel—has no mechanical advantage, because the rope is continuous. But by passing the rope underneath a pulley, the load is shared between two sections of the rope, and the effort is reduced by half; in that case, the load moves half as far as the end of the rope is pulled. By combining two or more pulley blocks, the mechanical advantage can be increased further.

SINGLE PULLEY

A rope passed over one pulley can raise a weight attached to the end of the rope. This set-up has no mechanical advantage, but it does change the direction of the force—and it can be more convenient than simply lifting the weight.

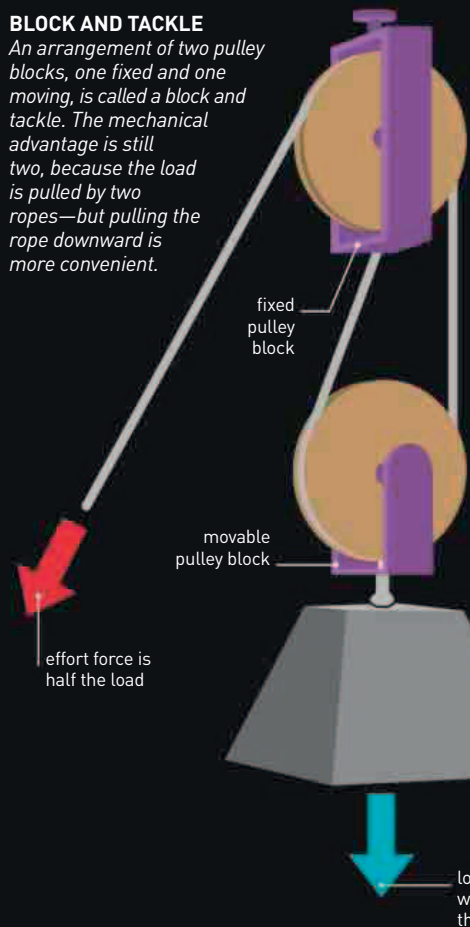


HALF THE EFFORT

A single pulley block can be used to create a mechanical advantage of two. If the rope is slung under the pulley wheel, the force is shared between the two sections of rope either side of the pulley.

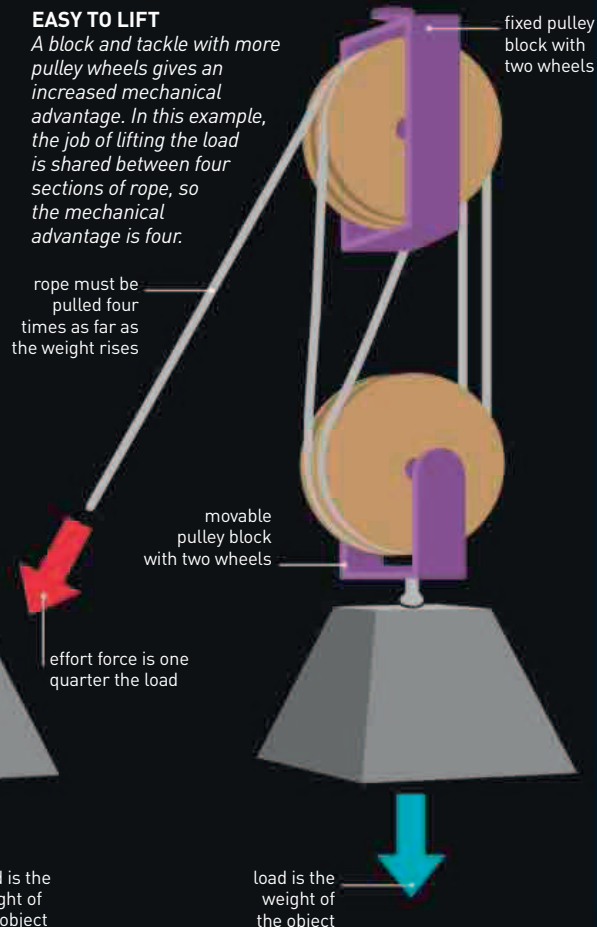
BLOCK AND TACKLE

An arrangement of two pulley blocks, one fixed and one moving, is called a block and tackle. The mechanical advantage is still two, because the load is pulled by two ropes—but pulling the rope downward is more convenient.



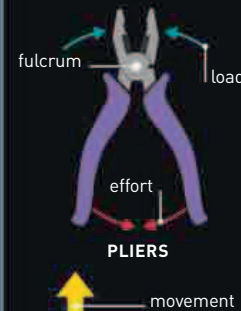
EASY TO LIFT

A block and tackle with more pulley wheels gives an increased mechanical advantage. In this example, the job of lifting the load is shared between four sections of rope, so the mechanical advantage is four.



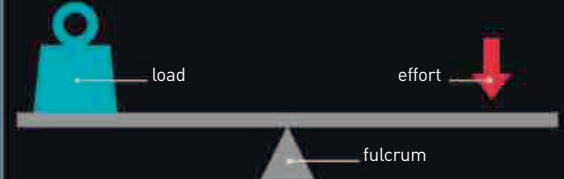
LEVERS

The mechanical advantage of a lever is the ratio of distances from the fulcrum (pivot) to the effort and the load. The ratio can be equal to one, or greater than or less than one. There are three types of lever, distinguished by the positions of the effort and load relative to the fulcrum.



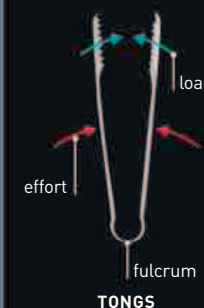
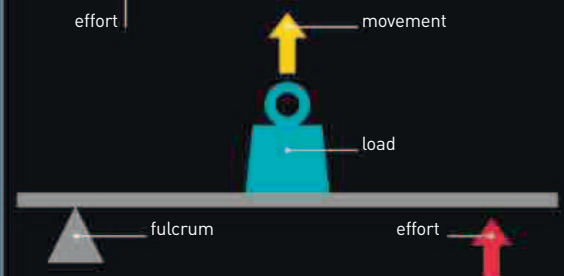
CLASS 1 LEVER

A class 1 lever has the fulcrum between the effort and the load. A seesaw is a familiar example—normally, the pivot is in the center with the load and effort at an equal distance, so there is no mechanical advantage. However, an adult sitting close to the pivot can be lifted by a child sitting on the opposite end.



CLASS 2 LEVER

In a class 2 lever, the load is closer to the fulcrum than the effort is—so the mechanical advantage is greater than one. This can make it easy to lift heavy weights, as in a wheelbarrow.



CLASS 3 LEVER

In a class 3 lever, the effort is closer to the fulcrum than the load, so the mechanical advantage is always less than one. The load moves farther (and faster) than the effort; a golf club benefits from this effect.

